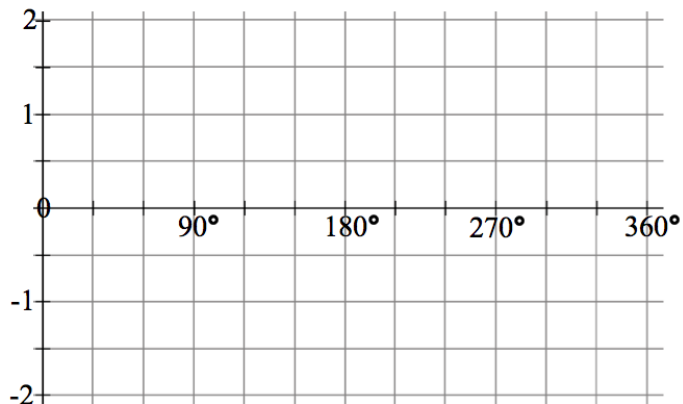


Graphing Sine and Cosine Functions Worksheet

MCR3U

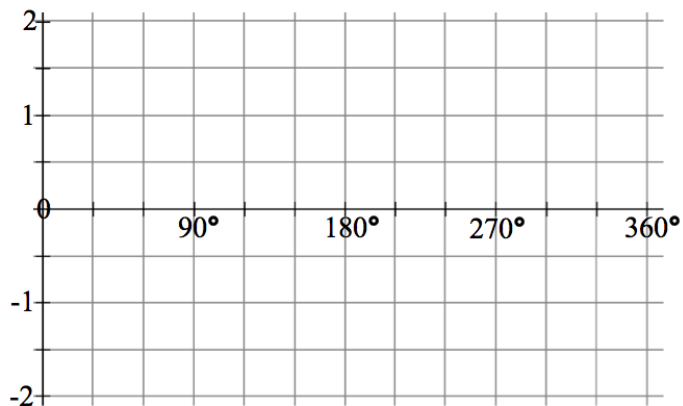
1) Graph the function $y = \sin x$ using key points between 0° and 360° .

x	y



2) Graph the function $y = \cos x$ using key points between 0° and 360° .

x	y



3) Determine the phase shift and the vertical shift of $y = \sin x$.

a) $y = \sin(x - 50^\circ) + 3$

b) $y = 2 \sin(x + 45^\circ) - 1$

4) Determine the phase shift and the vertical shift of $y = \cos x$.

a) $y = -9 \cos(x + 120^\circ) - 5$

b) $y = 12 \cos[5(x - 150^\circ)] + 7$

5) Determine the amplitude, the period, phase shift, vertical shift, maximum and minimum for each of the following.

a) $y = 5 \sin[4(x + 60^\circ)] - 2$

b) $y = 2 \cos[2(x + 150^\circ)] - 5$

c) $y = \frac{1}{2} \sin[\frac{1}{2}(x - 60^\circ)] + 1$

d) $y = 0.8 \cos[3.6(x - 40^\circ)] - 0.4$